

On the Origin of Stochastic Photometric Variability in Massive Stars

MATTEO CANTIELLO,^{1,2} DANIEL LECOANET,^{3,4,2} AND ADAM S. JERMYN¹

¹*Center for Computational Astrophysics, Flatiron Institute, New York, NY 10010, USA*

²*Department of Astrophysical Sciences, Princeton University, Princeton, NJ 08544, USA*

³*Department of Engineering Sciences & Applied Mathematics, Northwestern University, Evanston, IL 60201, USA*

⁴*CIERA, Northwestern University, Evanston, IL 60201, USA*

Submitted to ApJ

ABSTRACT

High-precision photometric observations have revealed a new ubiquitous phenomena in early type stars: stochastic low-frequency photometric variability. Possible explanations for the origin of this photometric variability include internal gravity waves launched by the convective core, or the presence of subsurface convection. Here we show theoretical predictions for the trend of characteristic frequency and amplitude of the perturbation induced by subsurface convection in massive stars, and compare these predictions to observations. We found that both the characteristic frequency ν_{char} and amplitude α_0 of the observed photometric variability correlate very well with properties in the iron subsurface convection zone. For the same stars we also confirm that the amplitude of macroturbulence tracks the convective velocity in the iron convection zone. We suggest a unified picture in which low-frequency photometric variability, surface turbulence, magnetic spots and wind variability in massive stars could all have the same origin: the presence of subsurface convection zones.

Keywords: convection, stars: magnetic field, stars: variability

1. INTRODUCTION

Massive stars largely drive the dynamical and chemical evolution of gas in galaxies. They accomplish this via their stellar winds, eruptions, and explosive deaths. The legacy of massive stars is continued by the compact remnants they leave behind, neutron stars and black holes. These remnants can merge and generate the gravitational waves observed by LIGO/Virgo. The evolutionary trajectory starting with a massive star burning hydrogen in its core and ending with a compact remnant is understood only qualitatively. We still do not know how to map initial properties of the star, like mass, rotation rate, and metallicity, to e.g. the final mass and spin of the compact remnant it leaves behind. The picture is actually even more complicated, as the majority of massive stars is found in multiple systems (Sana et al. 2012), with a large fraction expected to interact with a companion during its lifetime (de Mink et al. 2014).

The path towards a detailed understanding of massive stars begins with a quantitative study of their internal structure on the main sequence. In recent years asteroseismology has opened a new window to the interior of these challenging astrophysical environments, with high precision photometry from space delivering many new exciting results (e.g., MOST, CoRoT, BRITE, Kepler/K2 and TESS, see Bowman 2020). The latest discovery is the detection of a new ubiquitous phenomenon in massive stars, stochastic low-frequency photometric variability (Bowman et al. 2019a,b, 2020). This phenomenon joins a number of other surface and wind phenomena that are routinely observed in early-type stars, including surface turbulence (Simón-Díaz & Herrero 2014), line profile variability (Fullerton et al. 1996), and discrete absorption components in UV spectra (Howarth & Prinja 1989; Cranmer & Owocki 1996; Fullerton et al. 1997; Kaper et al. 1997). Other phenomena that are harder to observe but could still be common in these intrinsically bright stars are surface magnetism and bright spots (e.g. Ramiaramanantsoa et al. 2014).

The origin of this stochastic photometric variability is currently debated. It could be caused by internal gravity waves (IGWs) launched by turbulent core convection (Edelmann et al. 2019; Ratnasingam et al. 2020), or by sub-surface convection zones (Cantiello et al. 2009; Blomme et al. 2011; Lecoanet et al. 2019). Regardless of its origin, this photometric signal is likely carrying important information about the stellar structure, complementing asteroseismic studies that use well-identified oscillation modes (e.g. Aerts 2019; Aerts et al. 2019; Bowman 2020).

A recent development was the use of high resolution ground-based spectroscopy for the targets observed by K2 and TESS (Bowman et al. 2020). This allowed precise determination of stellar parameters, including mass, luminosity and a measure of surface velocity fluctuations (macroturbulence). Here we used this data to compare the observed properties of the stochastic photometric variability with some predictions coming from 1D stellar models. We discuss the properties of subsurface convection, in particular the expected trends in amplitude and frequency of the variability they could induce at the stellar surface, as a function of the stellar temperature and luminosity. We show that the predicted trends coming from a subsurface convection zone driven by the iron opacity peak at $\approx 150kK$ match very well the observations.

The mechanism exciting surface turbulent velocity fluctuations is likely the same that produces the observed stochastic photometric variability (Bowman et al. 2020). We show that massive stars with surface magnetic fields can be used to break the degeneracy between the two proposed mechanisms, with core IGWs being much more susceptible to the presence of a B-field. Macroturbulence is observed in stars with fairly strong magnetic fields, sufficient to suppress IGWs from the core. On the other hands, the only stars consistent with having no macroturbulence are stars where the magnetic field is strong enough to shut off the FeCZ, consistent with a subsurface origin of surface perturbations.

We conclude suggesting that subsurface convection represents a possible unifying mechanism for low-frequency photometric variability, surface turbulence, magnetic spots and wind variability in massive stars.

2. METHODS

We calculated stellar evolution models using the Modules for Experiments in Stellar Astrophysics (MESA Paxton et al. 2011, 2013, 2015, 2018, 2019) software instrument. Details on the microphysics inputs to this software instrument are given in Appendix A. The inlists used to run our models can be found in this [GitHub](#)

[repository](#)¹. We focused on the structure and convective properties of our models, in order to calculate typical frequencies and convective fluxes. Convection is calculated in the framework of the Mixing Length Theory (Böhm-Vitense 1958, MLT hereafter), and we adopted $\alpha_{\text{MLT}} = 1.6$. While in efficient convection zones (e.g. stellar cores) the choice of this parameter does not change results much, this is not true in superadiabatic, inefficient convective regions close to the stellar surface (Jiang et al. 2015; Cantiello & Braithwaite 2019). We discuss in Section 4 how uncertainties in the choice of the α_{MLT} parameter affect our results.

Since we are interested in the excitation of surface phenomena, we focus on the upper part of convective zones and following Cantiello et al. (2009) we define the average of a quantity $\bar{q}$ as

$$\bar{q} \equiv \frac{1}{\alpha_{\text{MLT}} \bar{H}_{\text{P}}} \int_{R_{\text{c}} - \alpha_{\text{MLT}} \bar{H}_{\text{P}}}^{R_{\text{c}}} q(r) dr \quad (1)$$

where H_{P} is the pressure scale height calculated at the upper boundary (R_{c}) of the convective zone of interest. We tested using different average procedures, and found that our results do not depend much on the specific choice.

Using this definition, we extracted the average Mach number $\bar{M}_{\text{c}}$, convective velocity $\bar{v}_{\text{c}}$ and density $\bar{\rho}_{\text{c}}$ in the convective core and in the subsurface convection zones of our models. The most important subsurface convection zone for the massive stars discussed in this work is the iron convection zone (FeCZ) (Cantiello et al. 2009), although in the low luminosity regime He-driven convection zones could play a role as well (Cantiello & Braithwaite 2019). We define the characteristic frequency as

$$\nu_{\text{char}} \equiv \frac{1}{2\pi\tau}, \quad (2)$$

where τ is a characteristic timescale. When calculating the characteristic frequency in our models, we set τ to be the average convective turnover time

$$\tau_{\text{c}} = \alpha_{\text{MLT}} \bar{H}_{\text{P}} / \bar{v}_{\text{c}} \quad (3)$$

calculated either in the FeCZ or in the convective core. We proceeded to calculate convective turnover timescales and other relevant properties like convective fluxes $F_{\text{c}} = \bar{\rho}_{\text{c}} \bar{v}_{\text{c}}^3$. We did this as a function of both mass and evolutionary history for stars ranging from $5 - 120 M_{\odot}$. Here we present results for an initial metallicity of $Z=0.02$, but in Appendix C we report results for model grids with $Z = 0.014, 0.006$ and 0.002 as well.

¹ <https://github.com/matteocantiello/rednoise>

We compare these quantities with the observed characteristic frequency ν_{char} and amplitude α_0 of stochastic photometric variability. These quantities are derived by fitting the stochastically-excited low-frequency power excess in a power density spectrum using a Lorentzian function (e.g. Bowman et al. 2019b):

$$\alpha(\nu) = \frac{\alpha_0}{1 + \left(\frac{\nu}{\nu_{\text{char}}}\right)^\gamma} + P_W. \quad (4)$$

This shows that α_0 represents the amplitude at zero frequency. ν_{char} is defined as in Eq. 2 with τ the typical timescale of the stochastic photometric variability. Finally, γ is the gradient of the linear part of the profile and P_W is a white noise term. Since the stochastic photometric variability is usually observed at low-frequencies, in the literature this is also referred to as “red noise”. We will use these terms interchangeably in the rest of the paper.

3. RESULTS

We want to test a possible correlation between the properties of the FeCZ and observed photospheric phenomena, in particular photometric variability and turbulent velocities at the stellar surface (macroturbulence). If the stochastic photometric variability was caused by the FeCZ, a natural choice of proxy for the typical timescale of this variability is the convective turnover timescale. We found typical values of the convective turnover timescale to be about $\sim 0.1 \dots 2$ d in the FeCZ, with a tendency for shorter values in models with high effective temperature and surface luminosity (Cantiello et al. 2009). To quantitatively compare to the characteristic frequencies of the observed stochastic photometric variability in massive stars, we proceed mapping our theoretical predictions for the quantity ν_{char} (eq. 2) on a spectroscopic H-R diagram (Langer & Kudritzki 2014). This reveals a very good agreement, with our models reproducing both the typical values of the observed ν_{char} and the trend with $\log \mathcal{L}$ and $\log T_{\text{eff}}$ (Fig. 1). While the characteristic frequencies found by Bowman et al. (2020) seem to be larger by a factor of ≈ 3 , our predictions for the turnover timescale are affected by uncertainty in the convective velocities arising from the MLT treatment ($v_c \propto \alpha_{\text{MLT}}^3$, so $\nu_{\text{char}} \propto \alpha_{\text{MLT}}^2$). Since α_{MLT} is uncertain by a factor of 2 or so, our estimates of the characteristic frequency in the FeCZ are uncertain by a factor of ≈ 4 .

We proceed with mapping quantities that measure the amplitude of perturbations at the location of the FeCZ. We then check if some of these properties correlate with the amplitude of observed photospheric phenomena, including turbulent velocity fluctuations and photometric

variability. While here we are agnostic about the particular mechanism, we assume that some physical process connects the subsurface layers to the photosphere. We discuss the most likely candidates for this process in Sec. 4. We show the average convective velocity in the FeCZ in Fig. 2. Velocities of the order $10 \dots 100 \text{ km s}^{-1}$ are found across the upper H-R diagram, with a trend of increasing v_c for higher luminosities. This trend is similar to what is observed for the macroturbulent velocity field at the surface of massive stars. This is in agreement with the results of Grassitelli et al. (2015b), who found a tight correlation between the turbulent pressure ($P_{\text{turb}} \propto v_c^2$) in the FeCZ and spectroscopically-derived macroturbulent velocities in massive stars. Interestingly, the range of the velocity field in the FeCZ is also similar to the observed range of macroturbulent velocities. Again the caveat is the large uncertainty (a factor ≈ 8) of the convective velocities calculated using the MLT in inefficient convective zones (Cantiello & Braithwaite 2019). We plot the average Mach number $\bar{M}_c$ of convection in the FeCZ in Fig. 6, showing that contrary to core convection the FeCZ is just mildly subsonic, with values of the Mach number ranging from 0.01 to 0.3. The relative convective flux in these regions is shown in Fig. 1, where we also report the observed amplitude of photometric variability. Values of F_c/F_* tend to increase with increasing $\log \mathcal{L}$ and decreasing $\log T_{\text{eff}}$, a trend that is also found for the values of observed photometric variability. The properties of the FeCZ correlate very well with the timescale and amplitude of the observed stochastic photometric variability, as well as the amplitude of macroturbulence.

3.1. Core Convection and Photometric Variability

[Daniel, this might need your touch]. Early-type stars possess convective cores during their main sequence. Turbulent convection in these convective cores excites internal gravity waves (Goldreich & Kumar 1990; Rogers et al. 2013; Lecoanet & Quataert 2013; Shiode et al. 2013; Edelmann et al. 2019; Horst et al. 2020), that can propagate through the stellar envelope and reach the stellar surface (e.g. Ratnasingam et al. 2019; Lecoanet et al. 2019; Ratnasingam et al. 2020). These waves could in principle be responsible for both the observed macroturbulence and photometric variability in early-type stars (Aerts et al. 2009; Simón-Díaz et al. 2017; Bowman et al. 2019b,a). Shiode et al. (2013) estimated that IGWs propagating from the core should lead to surface brightness fluctuations with typical amplitudes $\approx 10^{-2} - 10^2 \mu\text{mag}$, at odds with the large amplitudes $\approx 10 - 10^4 \mu\text{mag}$ observed by e.g. Bowman et al. (2020). Lecoanet et al. (2019) also argue that the low-frequency,

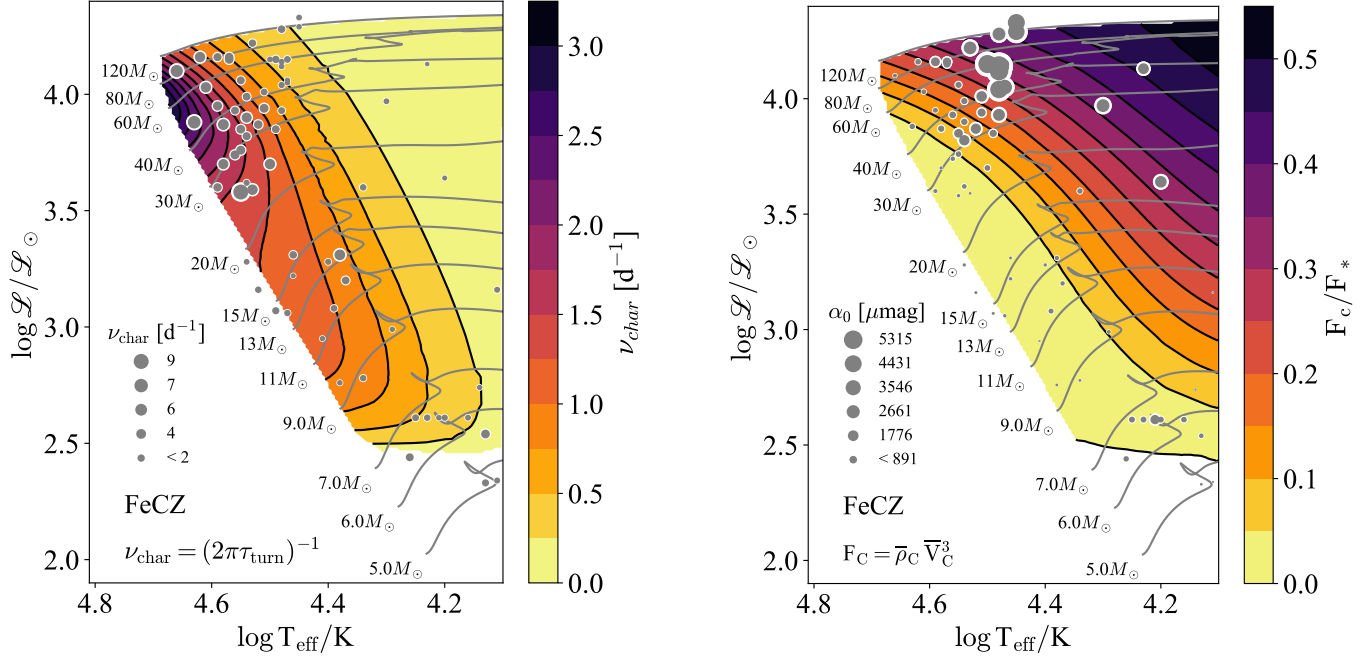


Figure 1. Left panel: Characteristic frequency ν_{char} in the FeCZ as a function of the location of stellar models in the spectroscopic H-R Diagram. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} . Right panel: Ratio of FeCZ convective flux to the total stellar flux in the spectroscopic H-R Diagram. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

stochastic variability is unlikely to be caused by linearly propagating waves from the core. Their predicted wave transfer function shows little power at very low frequencies due to radiative damping, and standing modes at frequencies above 0.5 d^{-1} . None of these features are present in the observed spectra. Recent numerical simulations of wave generation by convection in a $3M_{\odot}$ star (Edelmann et al. 2019; Horst et al. 2020) produce wave fluctuation spectra which are qualitatively similar to those observed. However, those simulations artificially boost the stellar luminosity by factors ranging from 10^3 to 10^7 . This makes it difficult to compare the amplitudes of the waves in those simulations to, e.g., Shiode et al. (2013), which predicts the surface variability in stars with realistic luminosities. **[Daniel: I don't think we need any discussion of wave spectra here... it's not used in our analysis.]**

Although the physics of wave generation by convection is still uncertain, here we analyze the core convective properties of our models. If the surface variability is due to core convection, one would expect the characteristic frequency and amplitude of the red noise to correlate with the properties of the core convection. The flux of IGWs excited by turbulent convection at the core boundary is $F_{\text{IGW}} = F_* \bar{M}_c$ (Goldreich & Kumar 1990). We expect the luminosity fluctuations to be

proportional to the relative surface radial displacement ξ_r/R produced by these waves at the stellar surface (e.g. Dziembowski 1977; Aerts et al. 2010). It can be shown that $\xi_r/R \propto \sqrt{F_{\text{IGW}}/F_0}$ (see Appendix B). This quantity is presented in Fig. 3. One important caveat is that this estimate neglects the important role of radiative damping, which is essential in shaping both the amplitude and the shape of the spectrum of waves at the surface (e.g. Lecoanet et al. 2019). The Mach number in the convective core is on the order of $10^{-3} \dots 10^{-4}$, with a trend of increasing values for lower $\log T_{\text{eff}}$ and higher $\log L$. Despite neglecting the important effect of radiative damping, the trend for the relative radial displacement at the surface do show some interesting correlations with the observed trend in photometric variability (right panel in Fig. 3).

Next, we focus on the characteristic timescale of waves excited by core convection. We expect the maximum of the IGW flux to be launched at frequencies close to $\nu_{\text{char}} = (2\pi\tau_c)^{-1}$, with τ_c defined in Eq. 3. Fig. 3 shows that the predicted values of ν_{char} are in the range $0.02 \dots 0.008 \text{ d}^{-1}$ ($0.2 \dots 0.08 \mu\text{Hz}$), in agreement with the results of Shiode et al. (2013). These values are about 2 orders of magnitude smaller than the typical characteristic frequencies of stochastic photometric variability observed by (Bowman et al. 2019b,a). This is

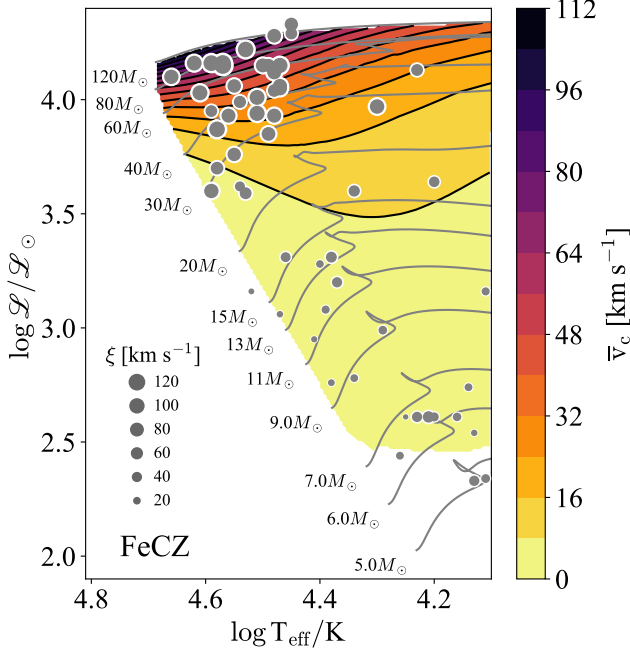


Figure 2. Average convective velocities in the FeCZ as function of the location of stellar models in the spectroscopic H-R Diagram. We also show the observed stars with detected macroturbulent velocity as grey dots. The size of the dot is proportional to the observed macroturbulence.

because wave propagation through the stellar envelope affects the spectrum, changing the frequency of maximum power of waves reaching the surface (Lecoanet et al. 2019). Radiative damping efficiently suppresses low-frequency IGWs (because these waves have high radial wavenumbers), so that the the peak of the spectrum observed at the surface is expected to move to higher frequencies. The amplitude of this effect depends on the envelope properties, which are a function of $\log T_{\text{eff}}$ and $\log \mathcal{L}$. Nevertheless, as a first approximation, we will assume that ν_{char} , as function of $\log T_{\text{eff}}$ and $\log \mathcal{L}$, is still set by the core convective properties. Under this assumption, Fig. 3 shows that ν_{char} should increase with both $\log T_{\text{eff}}$ and $\log \mathcal{L}$, so that the characteristic frequencies of peak IGWs should increase as stars evolve on the main sequence. This is exactly the opposite of what is observed: the characteristic frequencies of stochastic photometric variability is largest for stars early on on the main sequence, and seem to decreases as stars evolve. Therefore either radiative damping is able to revert this trend, or the observed variability might not be caused by IGWs launched by the core.

3.2. Macroturbulence

The hypothesis that subsurface convection is responsible for surface turbulence is corroborated by the match between trends in the FeCZ properties (i.e. convective velocities and turbulent pressure) and the observed micro and macroturbulence (Cantiello et al. 2009; Grassitelli et al. 2015b). One important test to this hypothesis is provided by the disappearance of macroturbulence in stars with surface magnetic fields above a critical strength (Sundqvist et al. 2013), closely corresponding to the critical field needed to stabilize the FeCZ (MacDonald & Petit 2019; Jermyn & Cantiello 2020). Note that in OB stars, the FeCZ is deeper and more vigorous than the H and He convection zones, so a magnetic field stabilizing the FeCZ will necessarily also stabilize the other subsurface convection zones.

We can analogously define a critical magnetic field strength B_{crit} which suffices to reflect IGWs before they reach the photosphere. Using the dispersion relation for IGWs in a magnetic medium, the radial component of this field is (Fuller et al. 2015)

$$B_{r,\text{crit}} = \frac{\omega}{2k_r} \sqrt{4\pi\rho}, \quad (5)$$

where $\omega = \nu/2\pi$ is the angular frequency of the waves, ρ is the density, and k_r is the radial wave-number. For IGWs the radial wave-number is related to the azimuthal quantum number ℓ by

$$k_r \approx \frac{\sqrt{\ell(\ell+1)}}{r} \left(\frac{N}{\omega} \right), \quad (6)$$

where N is the Brünt-Väisälä frequency. So

$$B_{r,\text{crit}} \approx \frac{\omega^2 r}{N} \sqrt{\frac{4\pi\rho}{2\ell(\ell+1)}} \quad (7)$$

$$\approx \frac{\omega^2 r}{\ell N} \sqrt{4\pi\rho}. \quad (8)$$

Note that because this decreases with increasing ℓ , all waves of a given frequency are reflected if that with $\ell = 1$ is reflected.

We computed $B_{r,\text{crit}}$ for several values of ℓ and ν for a main-sequence model of a $30M_{\odot}$ star as a function of radius, shown in Fig. 3.2. For frequencies comparable to those of core convection the critical magnetic field strength is of order 10^{-2} G to 10^{-1} G. Strong macroturbulence is observed in similar-mass O-type stars with magnetic fields up to 2.5 kG (e.g. HD 191612) (Sundqvist et al. 2013), so macroturbulence in those stars is unlikely to be due to IGW coming from their cores if those IGW have similar frequencies to that of core convection.

For frequencies comparable to the observed ν_{char} , on the order of 3 d^{-1} , the critical magnetic field is much

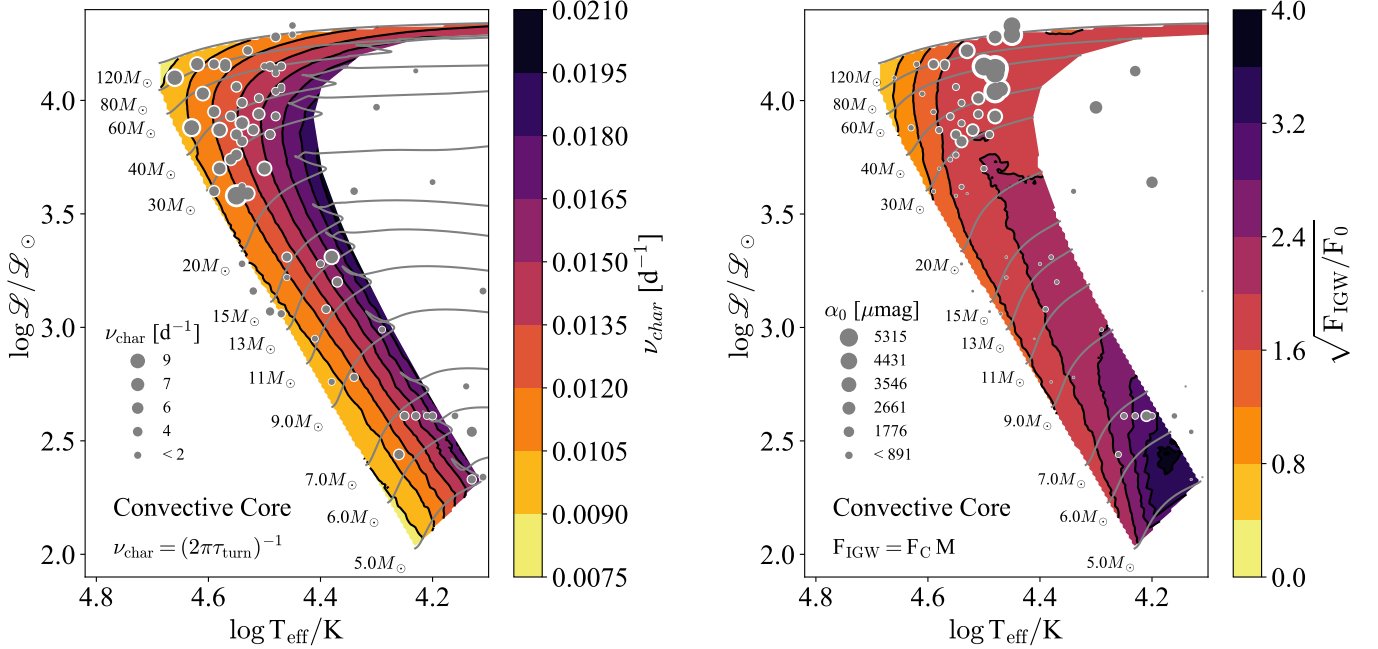


Figure 3. Left panel: Characteristic frequency ν_{char} in the convective core as function of the location of stellar models in the spectroscopic HRD. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} . Right panel: Square root of the ratio between the flux of gravity waves launched by the convective core and $F_0 = 1/2\rho(r)r^3\omega^2\sqrt{N(r)^2 - \omega^2}$, evaluated at the stellar surface. In the absence of damping, this quantity is expected to be proportional to the relative radial displacement at the surface (See Appendix B). The IGW flux is calculated multiplying the core convective flux by the average convective Mach number in the top pressure scale height of the convective region. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

larger, on the order of 300 G to 1 kG. This makes an explanation of macroturbulence based on IGW marginally inconsistent with observations showing strong macroturbulence up to field strengths of 2.5 kG. However, because equation (7) is a strong function of ω , and hence of ν , it is possible that these strongly magnetized stars just have larger ν_{char} . For instance if HD 191612 has $\nu_{\text{char}} \approx 5 \text{ d}^{-1}$ then it would have $B_{r,\text{crit}} \approx 3 \text{ kG}$ for the $\ell = 1$ mode.

A further prediction of this calculation is that, if IGW are the cause of macroturbulence in these stars, we should expect the strength of macroturbulence to decline with increasing magnetic field strength as more and more modes are reflected before they reach the surface. Sundqvist et al. (2013) find no such trend, and the only strong correlation they see of macroturbulence with magnetic field strength is that it vanishes when the magnetic field exceeds the FeCZ shutoff strength. This again points against an explanation of macroturbulence based on IGWs.

4. DISCUSSION

The presence of a subsurface convection zone can result in variability of photospheric properties via a num-

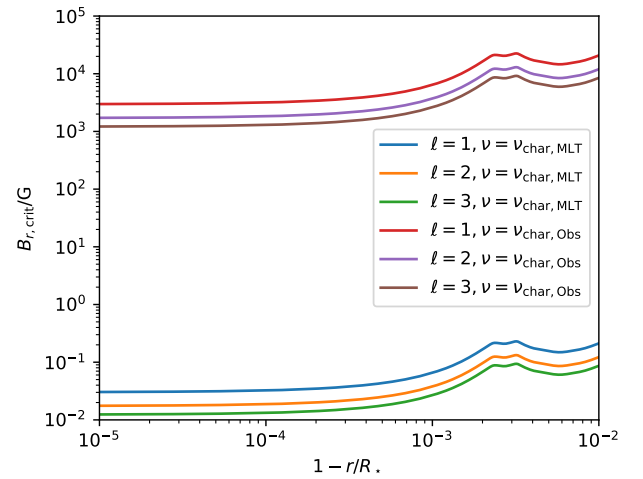


Figure 4. The critical radial magnetic field strength needed to reflect waves of different ℓ and ν is shown as a function of fractional depth for a $30M_\odot$ stellar model at an age of 4 Myr. The frequency $\nu_{\text{char,MLT}}$ is that given by equation (2) but evaluated for the core convection zone instead of the subsurface FeCZ. The frequency $\nu_{\text{char,Obs}}$ is a typical value of 3 d^{-1} for the observations.

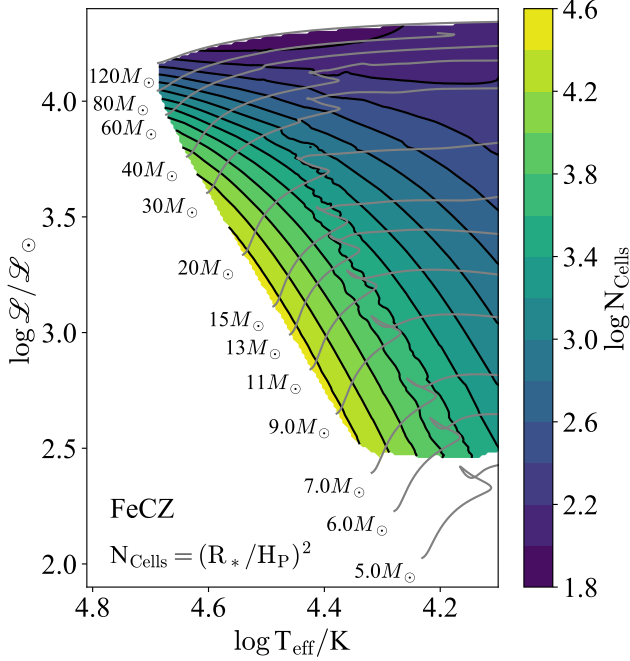


Figure 5. Expected number of convective cells in the FeCZ. This is calculated assuming the size of convective cells is about the average pressure scale height in the FeCZ

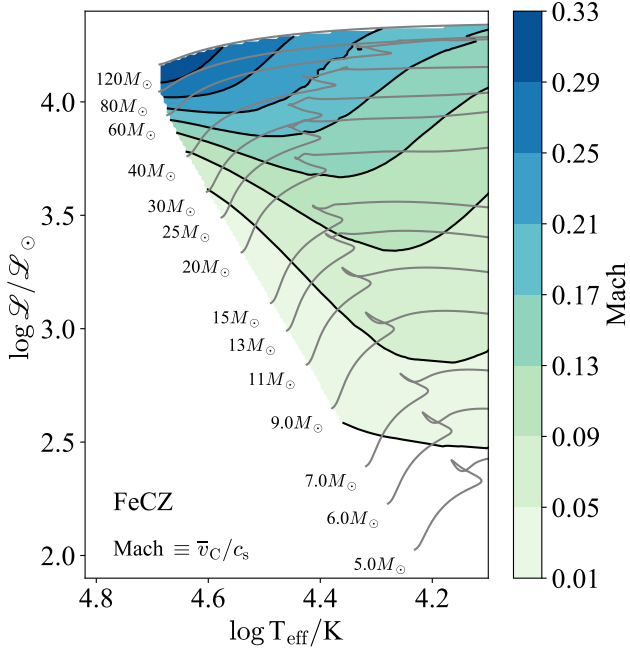


Figure 6. Mach number in the FeCZ.

ber of processes, including wave excitation (Cantiello et al. 2009; Grassitelli et al. 2015b) and magnetic buoyancy (Cantiello & Braithwaite 2011). Multi dimensional simulations including radiation have been performed in

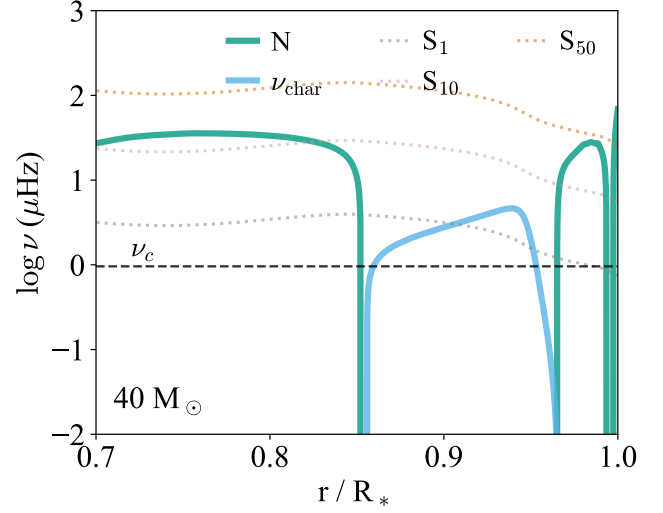


Figure 7. Propagation diagram for a $40M_{\odot}$ model showing the outer 30% of the star in radial coordinate.

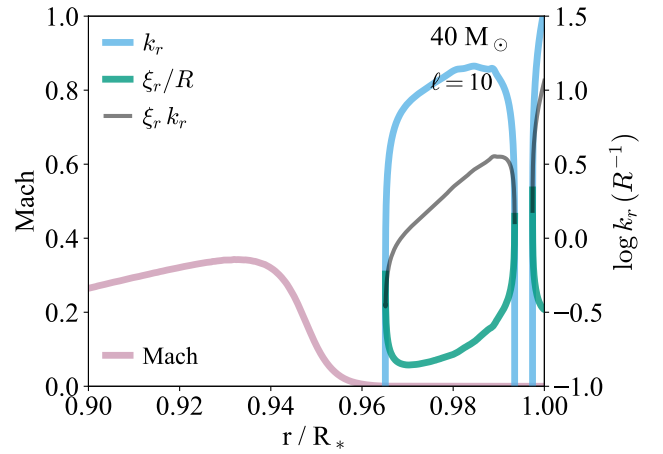


Figure 8. Properties of waves launched by the FeCZ in a $40M_{\odot}$ model. The plot shows the outer 10% of the star in radial coordinate. [MC: do we need this plot? Probably not if we do not discuss in detail IGW launched by the FeCZ. Which is probably better for this paper]

a restricted range of the parameter space, and show that the full turbulent manifestation of these convective regions extends up to the stellar surface (Jiang et al. 2015, 2017, 2018). In their calculations of OB stars envelopes including the stellar photosphere, Jiang et al. (2015) observe turbulent velocities reaching the isothermal sound speed ($\approx 50\text{km s}^{-1}$) at the stellar surface, demonstrating that velocity fields of amplitude comparable to the observed macroturbulence are naturally explained by the presence of the FeCZ.

Therefore it could be that the observed stochastic photometric variability and macroturbulence simply repre-

sent the direct manifestation of turbulent, radiation-dominated convection at the stellar surface. The simplified MLT treatment in our one dimensional calculations is unable to capture the complex phenomenology of these layers, but the fact that it can reproduce both the timescales and the trends in amplitude of the observed photometric variability is compelling. It calls for extending the radiation hydrodynamics simulations to cover the parameter space of the TESS observations, in order to unravel the precise mechanism connecting subsurface convection zones to the observed surface variability.

The results of (Bowman et al. 2019b) show that low-frequency stochastic photometric variability is also observed in low-metallicity, LMC stars. The presence of the FeCZ depends on the luminosity and metallicity of the star, and in 1D stellar evolution calculations the FeCZ occurs above a luminosity of $L \approx 10^{3.2} L_{\odot}$ for $Z=0.02$. This corresponds to a zero age main sequence star of about $7M_{\odot}$ (see e.g. Fig. 2). The LMC has a metallicity about half solar, and in models at $Z=0.008$ the FeCZ only appears at $L \approx 10^{3.9} L_{\odot}$, corresponding to a zero age main sequence star of about $11M_{\odot}$ (Cantiello et al. 2009). Note that these limits could move downward in the presence of atomic diffusion and radiative acceleration (Richer et al. 2000), or due to an upward revision of the uncertain values of Fe opacity (e.g. Bailey et al. 2015). It will be interesting to see if the TESS observations can probe the transition region between stars with and without a FeCZ, and determine a possible change in surface properties.

It is important to note that some low-luminosity, main sequence A stars in (Bowman et al. 2019b) show low-frequency, stochastic photometric variability. For these stars models do not predict the presence of the FeCZ. However, these stars still show (sub)surface convection zones triggered by ionization of H and He (Cantiello & Braithwaite 2019). The characteristic frequencies of convection in these regions are also in the range $\sim$ tens of μHz , although the amplitude of the velocity fluctuations they can induce is much smaller than for the FeCZ, so it is not clear if they could be linked to observed photometric variability and macroturbulence. This said it's clear that the regime of A and late-B type stars is the most favorable for the detection of core-generated IGWs. The relative perturbation induced by subsurface convection is weakest in this regime (Cantiello & Braithwaite 2019; Jermyn & Cantiello 2020), and depending on the amplitude of core generated IGWs it could well be subdominant, allowing for a detection. Shiode et al. (2013) predict that in this regime, the peak rms surface brightness fluctuations induced by core-generated IGWs should be in the range $\approx 0.01\text{--}100\mu\text{mag}$ (See their Fig. 6).

4.1. Red noise in evolved stars

Low-frequency, stochastic photometric variability with similar properties to the massive main sequence stars discussed by (Bowman et al. 2019b, 2020) was recently observed in evolved massive stars. Naze et al. (2021) detected red noise in both luminous blue variables (LBV) and Wolf-Rayet (WR) stars, and Dorn-Wallenstein et al. (2020) found the same photometric signature in yellow supergiants (YSG). Compared to OB stars, LBV, WR and YSG correspond to later stages of evolution. In particular, WR stars and YSG are likely burning helium in their cores. The internal structure of OB, LBV, WR and YSG stars changes dramatically, and this can affect substantially the generation and propagation of internal gravity waves. In particular, the amount of radiative damping is expected to change due to the sheer differences in envelope temperature and densities. We recall here that the radiative damping rate for a traveling g-mode is given by

$$\gamma_{\text{rad}}(\omega, \ell, r) = K_{\text{rad}}(r) k_r^2, \quad (9)$$

$$K_{\text{rad}}(r) = \frac{16 \sigma T(r)^3}{3 \rho(r)^2 \kappa(r) c_p(r)}, \quad (10)$$

$$(11)$$

where k_r is the radial wavenumber (eqn. 6) and κ is the opacity. Compact WR stars have surface temperatures that can exceed $\approx 10^5\text{K}$, while OB stars and LBVs have effective temperatures $\approx 10^{4.7\text{--}4}\text{K}$, with LBVs found at the cooler end of this range. The surface of extended, low-density, YSGs is cooler than 10^4K . With different types of core convection (H-burning vs He-burning) and radiative damping properties, it would be surprising if core-generated IGWs in e.g. OB and WR stars showed up at the surface with similar properties. On the other hand, the FeCZ is present below/at the surface in both OB and WR stars, and with similar velocities and convective turnover times. This seems to strengthen the main thesis of this work, adding support to a (sub)surface origin of the observed red noise.

The discussion is a bit more complicated for LBV and YSG, since at lower temperatures other convective regions can become prominent (driven by H and He recombination, see e.g. Jiang et al. 2018). We plan to systematically study the properties of (sub)surface convection in these evolved, cool stars in future work. Dorn-Wallenstein et al. (2020) ruled out a near-surface origin for the observed red noise in YSG, on the ground that the observed timescale do not follow the predicted scaling for granulation (Kallinger et al. 2014). The Kallinger scaling is applicable to granulation driven by surface convection, but is not appropriate for subsurface convec-

tion (see e.g. Fig.B.1 in Grassitelli et al. 2015a). This is because subsurface convective regions do not track the stellar surface properties, but move below the stellar surface to mass coordinates corresponding to their relevant ionization temperature (corresponding to an increase of the opacity and/or a decrease of the adiabatic gradient, see Cantiello & Braithwaite 2019). Therefore the argument of Dorn-Wallenstein et al. (2020) can not be used to rule out subsurface convection as a source of stochastic photometric variability (Lecoanet et al. 2019).

4.2. Towards a unified model for surface phenomena in massive stars

The presence of subsurface convection can simultaneously account for a large variety of puzzling phenomena that appear ubiquitous at the surface of early-type stars. First we have microturbulence and macroturbulence, and both can be accounted for by velocity fields excited by the underlying FeCZ subsurface convection zone (Cantiello et al. 2009; Grassitelli et al. 2015b). The only stars with no macroturbulence appear to have magnetic fields strong enough to shut off convection in the FeCZ, while stars with slightly lower surface magnetic fields show normal values of macroturbulence. Then we have the observation of bright spots in early-type stars (e.g. Ramiamananantsoa et al. 2014), which can be explained with the presence of magnetic spots rising from subsurface convective layers (Cantiello & Braithwaite 2011, 2019). These photospheric magnetic spots can then be responsible for another ubiquitous phenomena in early-type stars, the presence of discrete absorption components (DACs) in their UV spectra (e.g. Howarth & Prinja 1989; Cranmer & Owocki 1996; Fullerton et al. 1997; Kaper et al. 1997). Overall subsurface convection is expected to drive velocity and density perturbations at the stellar surface (e.g. Cantiello et al. 2009; Jiang et al. 2015). These perturbations could explain the ubiquitous line profile variability in hot stars (Fullerton et al. 1996) and provide a seed for the rapid development of wind instabilities that lead to wind clumping (Owocki et al. 1988), but see also (Sundqvist & Owocki 2013). Finally, these surface velocity and density perturbations would necessarily induce photometric variability, and we have shown here that the observed low-frequency stochastic noise detected in OB stars correlates very well with the presence and properties of the FeCZ.

An economical hypothesis emerges: the presence of subsurface convection, and in particular of the FeCZ,

could be the common underlying physical cause for the appearance of turbulence, magnetic spots, low-frequency photometric variability, as well as wind and spectroscopic variability in early-type stars.

5. CONCLUSIONS

We used one-dimensional stellar evolution calculations to study the predicted trends in the properties of subsurface convection in the spectroscopic HR-Diagram. We found that the trends of relative convective flux and convective turnover timescale in the FeCZ of our models match very well the trends in timescale and amplitude of low-frequency stochastic photometric variability in OB stars observed by TESS and K2.

Similar to previous works, we show that the observed trends in stellar macroturbulence are also well reproduced assuming the FeCZ as its driver. This connection is also supported by the observations of strongly magnetized early-type stars, showing no macroturbulence only for magnetic fields above the critical value required to shut-off turbulent convection in the FeCZ. We find that IGWs coming from the stellar core would be reflected or damped for values of the magnetic field well below this critical value. The fact that stars with strong but subcritical magnetic fields show typical values of macroturbulence points against a convective core origin of this surface perturbation.

Overall the observations support a picture in which subsurface convection, and in particular the FeCZ, is responsible for the ubiquitous low-frequency, stochastic photometric variability and macroturbulence detected in OB stars. These surface manifestations join a number of phenomena observed in early-type stars and attributed to the presence of subsurface convection, including the observations of (magnetic) bright spots as well as wind and spectroscopic variability. Radiation (magneto)hydrodynamics simulations of the outer envelope regions of early-type stars are required to understand the details of how subsurface convection zones cause the observed surface perturbations.

ACKNOWLEDGMENTS

The Center for Computational Astrophysics at the Flatiron Institute is supported by the Simons Foundation. This research was supported in part by the National Science Foundation under Grant No. NSF PHY-1748958 and by the Gordon and Betty Moore Foundation through Grant GBMF7392

APPENDIX

A. SOFTWARE DETAILS

Calculations were done with **MESA** version 11701. The **MESA** EOS is a blend of the OPAL Rogers & Nayfonov (2002), SCVH Saumon et al. (1995), PTEH Pols et al. (1995), HELM Timmes & Swesty (2000), and PC Potekhin & Chabrier (2010) EOSes. Radiative opacities are primarily from OPAL (Iglesias & Rogers 1993, 1996), with low-temperature data from Ferguson et al. (2005) and the high-temperature, Compton-scattering dominated regime by Buchler & Yueh (1976). Electron conduction opacities are from Cassisi et al. (2007). Nuclear reaction rates are a combination of rates from NACRE (Angulo et al. 1999), JINA REACLIB (Cyburt et al. 2010), plus additional tabulated weak reaction rates Fuller et al. (1985); Oda et al. (1994); Langanke & Martínez-Pinedo (2000). Screening is included via the prescription of Chugunov et al. (2007). Thermal neutrino loss rates are from Itoh et al. (1996). The inlists, processing scripts, and model output are available at Zenodo.org.

B. RADIAL DISPLACEMENT FROM IGWS

Let's consider traveling internal gravity waves in the WKB approximation. The wave flux is $F_{\text{IGW}} = \frac{1}{2}\rho|u|^2u_{g,r}$ with the group velocity $u_{g,r} = \partial\omega/\partial\kappa$. Since $u_{g,r} = (\omega/N)^2\sqrt{N^2 - \omega^2}/k_h$, $|u|^2 = (N/\omega)^2u_r^2$, and $u_r = \omega\xi_r$ we obtain:

$$F_{\text{IGW}} = \frac{1}{2}\rho\xi_r^2\omega^2\frac{\sqrt{N^2 - \omega^2}}{k_h}. \quad (\text{B1})$$

Using $k_h = \sqrt{\ell(\ell+1)}/r$ we can then write the relative radial displacement as:

$$\frac{\xi_r}{r} = \left[\frac{F_{\text{IGW}}\sqrt{\ell(\ell+1)}}{1/2\rho r^3\omega^2\sqrt{N^2 - \omega^2}} \right]^{1/2} \propto \sqrt{\frac{F_{\text{IGW}}}{F_0}} \quad (\text{B2})$$

C. GRIDS AT DIFFERENT METALLICITIES

Here we present results for the critical and surface magnetic field at metallicities of $Z=0.014$, $Z=0.006$, and $Z=0.002$, representing early-type stars in the Galaxy (MW), Large Magellanic Cloud (LMC), and Small Magellanic Cloud (SMC) respectively (e.g. Yusof et al. 2013). The initial Helium content of the grids is $Y=0.2659$ (MW), $Y=0.2559$ (LMC), and $Y=0.2508$ (SMC). The metallicity is initialized scaling the standard solar composition of Grevesse & Sauval (1998).

REFERENCES

- Aerts, C. 2019, arXiv e-prints, arXiv:1912.12300.
<https://arxiv.org/abs/1912.12300>
- Aerts, C., Christensen-Dalsgaard, J., & Kurtz, D. W. 2010, *Asteroseismology*
- Aerts, C., Mathis, S., & Rogers, T. M. 2019, *ARA&A*, 57, 35, doi: [10.1146/annurev-astro-091918-104359](https://doi.org/10.1146/annurev-astro-091918-104359)
- Aerts, C., Puls, J., Godart, M., & Dupret, M. A. 2009, *A&A*, 508, 409, doi: [10.1051/0004-6361/200810471](https://doi.org/10.1051/0004-6361/200810471)
- Angulo, C., Arnould, M., Rayet, M., et al. 1999, *Nuclear Physics A*, 656, 3, doi: [10.1016/S0375-9474\(99\)00030-5](https://doi.org/10.1016/S0375-9474(99)00030-5)
- Bailey, J. E., Nagayama, T., Loisel, G. P., et al. 2015, *Nature*, 517, 56, doi: [10.1038/nature14048](https://doi.org/10.1038/nature14048)
- Blomme, R., Mahy, L., Catala, C., et al. 2011, *A&A*, 533, A4, doi: [10.1051/0004-6361/201116949](https://doi.org/10.1051/0004-6361/201116949)
- Böhm-Vitense, E. 1958, *ZA*, 46, 108
- Bowman, D. M. 2020, arXiv e-prints, arXiv:2008.11162.
<https://arxiv.org/abs/2008.11162>
- Bowman, D. M., Burssens, S., Simón-Díaz, S., et al. 2020, *A&A*, 640, A36, doi: [10.1051/0004-6361/202038224](https://doi.org/10.1051/0004-6361/202038224)
- Bowman, D. M., Burssens, S., Pedersen, M. G., et al. 2019a, *Nature Astronomy*, 3, 760, doi: [10.1038/s41550-019-0768-1](https://doi.org/10.1038/s41550-019-0768-1)
- Bowman, D. M., Aerts, C., Johnston, C., et al. 2019b, *A&A*, 621, A135, doi: [10.1051/0004-6361/201833662](https://doi.org/10.1051/0004-6361/201833662)
- Buchler, J. R., & Yueh, W. R. 1976, *ApJ*, 210, 440, doi: [10.1086/154847](https://doi.org/10.1086/154847)
- Cantiello, M., & Braithwaite, J. 2011, *A&A*, 534, A140, doi: [10.1051/0004-6361/201117512](https://doi.org/10.1051/0004-6361/201117512)
- . 2019, *ApJ*, 883, 106, doi: [10.3847/1538-4357/ab3924](https://doi.org/10.3847/1538-4357/ab3924)
- Cantiello, M., Langer, N., Brott, I., et al. 2009, *A&A*, 499, 279, doi: [10.1051/0004-6361/200911643](https://doi.org/10.1051/0004-6361/200911643)
- Cassisi, S., Potekhin, A. Y., Pietrinferni, A., Catelan, M., & Salaris, M. 2007, *ApJ*, 661, 1094, doi: [10.1086/516819](https://doi.org/10.1086/516819)
- Chugunov, A. I., Dewitt, H. E., & Yakovlev, D. G. 2007, *PhRvD*, 76, 025028, doi: [10.1103/PhysRevD.76.025028](https://doi.org/10.1103/PhysRevD.76.025028)
- Couston, L.-A., Lecoanet, D., Favier, B., & Le Bars, M. 2018, *Journal of Fluid Mechanics*, 854, R3, doi: [10.1017/jfm.2018.669](https://doi.org/10.1017/jfm.2018.669)

- Cranmer, S. R., & Owocki, S. P. 1996, *ApJ*, 462, 469, doi: [10.1086/177166](https://doi.org/10.1086/177166)
- Cyburt, R. H., Amthor, A. M., Ferguson, R., et al. 2010, *ApJS*, 189, 240, doi: [10.1088/0067-0049/189/1/240](https://doi.org/10.1088/0067-0049/189/1/240)
- de Mink, S. E., Sana, H., Langer, N., Izzard, R. G., & Schneider, F. R. N. 2014, *ApJ*, 782, 7, doi: [10.1088/0004-637X/782/1/7](https://doi.org/10.1088/0004-637X/782/1/7)
- Dorn-Wallenstein, T. Z., Levesque, E. M., Neugent, K. F., et al. 2020, *ApJ*, 902, 24, doi: [10.3847/1538-4357/abb318](https://doi.org/10.3847/1538-4357/abb318)
- Dziembowski, W. 1977, *AcA*, 27, 203
- Edelmann, P. V. F., Ratnasingam, R. P., Pedersen, M. G., et al. 2019, *ApJ*, 876, 4, doi: [10.3847/1538-4357/ab12df](https://doi.org/10.3847/1538-4357/ab12df)
- Ferguson, J. W., Alexander, D. R., Allard, F., et al. 2005, *ApJ*, 623, 585, doi: [10.1086/428642](https://doi.org/10.1086/428642)
- Fuller, G. M., Fowler, W. A., & Newman, M. J. 1985, *ApJ*, 293, 1, doi: [10.1086/163208](https://doi.org/10.1086/163208)
- Fuller, J., Cantiello, M., Stello, D., Garcia, R. A., & Bildsten, L. 2015, *Science*, 350, 423, doi: [10.1126/science.aac6933](https://doi.org/10.1126/science.aac6933)
- Fullerton, A. W., Gies, D. R., & Bolton, C. T. 1996, *ApJS*, 103, 475, doi: [10.1086/192285](https://doi.org/10.1086/192285)
- Fullerton, A. W., Massa, D. L., Prinja, R. K., Owocki, S. P., & Cranmer, S. R. 1997, *A&A*, 327, 699
- Goldreich, P., & Kumar, P. 1990, *ApJ*, 363, 694, doi: [10.1086/169376](https://doi.org/10.1086/169376)
- Grassitelli, L., Fossati, L., Langer, N., et al. 2015a, *A&A*, 584, L2, doi: [10.1051/0004-6361/201527289](https://doi.org/10.1051/0004-6361/201527289)
- Grassitelli, L., Fossati, L., Simón-Díaz, S., et al. 2015b, *ApJL*, 808, L31, doi: [10.1088/2041-8205/808/1/L31](https://doi.org/10.1088/2041-8205/808/1/L31)
- Grevesse, N., & Sauval, A. J. 1998, *SSRv*, 85, 161, doi: [10.1023/A:1005161325181](https://doi.org/10.1023/A:1005161325181)
- Horst, L., Edelmann, P. V. F., Andrásy, R., et al. 2020, *A&A*, 641, A18, doi: [10.1051/0004-6361/202037531](https://doi.org/10.1051/0004-6361/202037531)
- Howarth, I. D., & Prinja, R. K. 1989, *ApJS*, 69, 527, doi: [10.1086/191321](https://doi.org/10.1086/191321)
- Iglesias, C. A., & Rogers, F. J. 1993, *ApJ*, 412, 752, doi: [10.1086/172958](https://doi.org/10.1086/172958)
- . 1996, *ApJ*, 464, 943, doi: [10.1086/177381](https://doi.org/10.1086/177381)
- Itoh, N., Hayashi, H., Nishikawa, A., & Kohyama, Y. 1996, *ApJS*, 102, 411, doi: [10.1086/192264](https://doi.org/10.1086/192264)
- Jermyn, A. S., & Cantiello, M. 2020, *ApJ*, 900, 113, doi: [10.3847/1538-4357/ab9e70](https://doi.org/10.3847/1538-4357/ab9e70)
- Jiang, Y.-F., Cantiello, M., Bildsten, L., Quataert, E., & Blaes, O. 2015, *ApJ*, 813, 74, doi: [10.1088/0004-637X/813/1/74](https://doi.org/10.1088/0004-637X/813/1/74)
- . 2017, *ApJ*, 843, 68, doi: [10.3847/1538-4357/aa77b0](https://doi.org/10.3847/1538-4357/aa77b0)
- Jiang, Y.-F., Cantiello, M., Bildsten, L., et al. 2018, *Nature*, 561, 498, doi: [10.1038/s41586-018-0525-0](https://doi.org/10.1038/s41586-018-0525-0)
- Kallinger, T., De Ridder, J., Hekker, S., et al. 2014, *A&A*, 570, A41, doi: [10.1051/0004-6361/201424313](https://doi.org/10.1051/0004-6361/201424313)
- Kaper, L., Henrichs, H. F., Fullerton, A. W., et al. 1997, *A&A*, 327, 281
- Langanke, K., & Martínez-Pinedo, G. 2000, *Nuclear Physics A*, 673, 481, doi: [10.1016/S0375-9474\(00\)00131-7](https://doi.org/10.1016/S0375-9474(00)00131-7)
- Langer, N., & Kudritzki, R. P. 2014, *A&A*, 564, A52, doi: [10.1051/0004-6361/201423374](https://doi.org/10.1051/0004-6361/201423374)
- Lecoanet, D., & Quataert, E. 2013, *MNRAS*, 430, 2363, doi: [10.1093/mnras/stt055](https://doi.org/10.1093/mnras/stt055)
- Lecoanet, D., Cantiello, M., Quataert, E., et al. 2019, *ApJL*, 886, L15, doi: [10.3847/2041-8213/ab5446](https://doi.org/10.3847/2041-8213/ab5446)
- MacDonald, J., & Petit, V. 2019, *MNRAS*, 487, 3904, doi: [10.1093/mnras/stz1545](https://doi.org/10.1093/mnras/stz1545)
- Naze, Y., Rauw, G., & Gosset, E. 2021, arXiv e-prints, arXiv:2101.05521. <https://arxiv.org/abs/2101.05521>
- Oda, T., Hino, M., Muto, K., Takahara, M., & Sato, K. 1994, *Atomic Data and Nuclear Data Tables*, 56, 231, doi: [10.1006/adnd.1994.1007](https://doi.org/10.1006/adnd.1994.1007)
- Owocki, S. P., Castor, J. I., & Rybicki, G. B. 1988, *ApJ*, 335, 914, doi: [10.1086/166977](https://doi.org/10.1086/166977)
- Paxton, B., Bildsten, L., Dotter, A., et al. 2011, *ApJS*, 192, 3, doi: [10.1088/0067-0049/192/1/3](https://doi.org/10.1088/0067-0049/192/1/3)
- Paxton, B., Cantiello, M., Arras, P., et al. 2013, *ApJS*, 208, 4, doi: [10.1088/0067-0049/208/1/4](https://doi.org/10.1088/0067-0049/208/1/4)
- Paxton, B., Marchant, P., Schwab, J., et al. 2015, *ApJS*, 220, 15, doi: [10.1088/0067-0049/220/1/15](https://doi.org/10.1088/0067-0049/220/1/15)
- Paxton, B., Schwab, J., Bauer, E. B., et al. 2018, *ApJS*, 234, 34, doi: [10.3847/1538-4365/aaa5a8](https://doi.org/10.3847/1538-4365/aaa5a8)
- Paxton, B., Smolec, R., Schwab, J., et al. 2019, *ApJS*, 243, 10, doi: [10.3847/1538-4365/ab2241](https://doi.org/10.3847/1538-4365/ab2241)
- Pols, O. R., Tout, C. A., Eggleton, P. P., & Han, Z. 1995, *MNRAS*, 274, 964, doi: [10.1093/mnras/274.3.964](https://doi.org/10.1093/mnras/274.3.964)
- Potekhin, A. Y., & Chabrier, G. 2010, *Contributions to Plasma Physics*, 50, 82, doi: [10.1002/ctpp.201010017](https://doi.org/10.1002/ctpp.201010017)
- Ramiamananantsoa, T., Moffat, A. F. J., Chené, A.-N., et al. 2014, *MNRAS*, 441, 910, doi: [10.1093/mnras/stu619](https://doi.org/10.1093/mnras/stu619)
- Ratnasingam, R. P., Edelmann, P. V. F., & Rogers, T. M. 2019, *MNRAS*, 482, 5500, doi: [10.1093/mnras/sty3086](https://doi.org/10.1093/mnras/sty3086)
- . 2020, *MNRAS*, doi: [10.1093/mnras/staa2296](https://doi.org/10.1093/mnras/staa2296)
- Richer, J., Michaud, G., & Turcotte, S. 2000, *ApJ*, 529, 338, doi: [10.1086/308274](https://doi.org/10.1086/308274)
- Rogers, F. J., & Nayfonov, A. 2002, *ApJ*, 576, 1064, doi: [10.1086/341894](https://doi.org/10.1086/341894)
- Rogers, T. M., Lin, D. N. C., McElwaine, J. N., & Lau, H. H. B. 2013, *ApJ*, 772, 21, doi: [10.1088/0004-637X/772/1/21](https://doi.org/10.1088/0004-637X/772/1/21)
- Sana, H., de Mink, S. E., de Koter, A., et al. 2012, *Science*, 337, 444, doi: [10.1126/science.1223344](https://doi.org/10.1126/science.1223344)
- Saumon, D., Chabrier, G., & van Horn, H. M. 1995, *ApJS*, 99, 713, doi: [10.1086/192204](https://doi.org/10.1086/192204)

- Shiode, J. H., Quataert, E., Cantiello, M., & Bildsten, L. 2013, MNRAS, 430, 1736, doi: [10.1093/mnras/sts719](https://doi.org/10.1093/mnras/sts719)
- Simón-Díaz, S., Godart, M., Castro, N., et al. 2017, A&A, 597, A22, doi: [10.1051/0004-6361/201628541](https://doi.org/10.1051/0004-6361/201628541)
- Simón-Díaz, S., & Herrero, A. 2014, A&A, 562, A135, doi: [10.1051/0004-6361/201322758](https://doi.org/10.1051/0004-6361/201322758)
- Sundqvist, J. O., & Owocki, S. P. 2013, MNRAS, 428, 1837, doi: [10.1093/mnras/sts165](https://doi.org/10.1093/mnras/sts165)
- Sundqvist, J. O., Petit, V., Owocki, S. P., et al. 2013, Monthly Notices of the Royal Astronomical Society, 433, 2497, doi: [10.1093/mnras/stt921](https://doi.org/10.1093/mnras/stt921)
- Timmes, F. X., & Swesty, F. D. 2000, ApJS, 126, 501, doi: [10.1086/313304](https://doi.org/10.1086/313304)
- Yusof, N., Hirschi, R., Meynet, G., et al. 2013, MNRAS, 433, 1114, doi: [10.1093/mnras/stt794](https://doi.org/10.1093/mnras/stt794)